Two Phase Interleaved Coupled Inductor Buck Converter Module

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Introduction

The design of high density, high efficiency DC-DC converters increasingly depends on the tight integration of electrical, thermal, and mechanical considerations within the power module itself. This project focuses on the development of a 72 V to 24 V, 1.1 kW, two phase interleaved coupled inductor buck converter operating at 200 kHz, following the architectural guidance of TI's TIDA-050089 reference design and the coupled inductor analysis presented by Jong Pil Lee et al. Unlike traditional PCB based converters, this module adopts a vertically stacked structure incorporating an Alpha Novatech heatsink, a TDK coupled inductor, and a central switching layer populated with EPC eGaN FETs, low profile decoupling capacitors, and high current terminals. The goal is to explore how physical layout, material selection, and parasitic behavior fundamentally shape converter performance, particularly in fast switching GaN systems where nanohenry level inductances and picofarad level capacitances directly influence voltage overshoot, switching losses, and EMI.

In this expanded phase of the project, the converter is further developed into a fully integrated power module by adding dedicated high side and low side gate driver circuits, gate drive capacitors, interstitial shielding layers for common mode noise reduction, and a complete mechanical housing. Although the assignment provides the option of a two sided cooling architecture using dielectric fluid, this design intentionally retains the single sided cooling approach, relying on the Alpha Novatech heatsink as the primary thermal extraction path. This choice preserves the original vertical stack topology, heatsink on one side, coupled inductor on the other, while avoiding the mechanical complexity of fluid channels and dual sided sealing, allowing the design effort to focus on electrical performance, parasitic minimization, and EMI control. This works out in our favor due to the particular EPC QFN package that we chose which has an exposed top cooling thermal pad that directly dissipates a majority of the heat through the heat sink without heating the entire board.

To meet the objectives of this project, the work integrates thermomechanical modeling, electrical loss analysis, 3D module layout, and ANSYS Q3D parasitic extraction into a cohesive design workflow. The first stage focuses on material behavior within the Cu/nano-Cu/ARC stack, establishing feasible conductor thicknesses, thermal limits, and mechanical stress margins that guide the physical construction of the module under single sided heatsink cooling. The subsequent stage develops the electrical and physical design of the two phase interleaved converter, ensuring that current paths, decoupling strategies, gate driver placement, shielding layer routing, and copper geometries satisfy both electrical performance targets and thermal constraints. Finally, parasitic inductances and capacitances extracted from the 3D layout are incorporated back into the SPICE simulations to evaluate their influence on switching node behavior, gate drive operation, and device stress. Collectively, these analyses provide a unified understanding of how packaging, materials, gate drive architecture, and layout decisions shape the real world performance of a high power GaN based converter module.

Problem Statement

What Is Given

The converter to be designed is a 72 V to 24 V, 1.1 kW, two-phase interleaved, coupled-inductor buck converter based on the architectural principles outlined in TI's TIDA-050089 reference design. The system employs a coupled inductor to reduce ripple current, improve transient performance, and enable current sharing between phases. Each phase uses EPC eGaN FETs, selected for their low switching loss and high-frequency capability, enabling efficient operation at the chosen 200 kHz switching frequency.

The power module is constructed on an ARC (Alumina Ribbon Ceramic) substrate, which serves as the primary structural and electrical interconnect platform. Copper and nano-copper layers form the metallization stack, providing low-resistance current paths and acting as thermal conduits to the external heatsink. The physical layout follows a vertically stacked structure with the Alpha Novatech heatsink on one side, the TDK coupled inductor on the other, and the switching devices and low-profile decoupling capacitors embedded in the central layer. This project retains the single-sided heatsink cooling approach, rather than the optional two-sided dielectric-fluid configuration.

In addition to the switching stage, the expanded assignment requires the integration of dedicated high-side and low-side gate-driver circuits, gate-drive capacitors, and voltage-supply terminals for the gate-drive network. The ARC substrate stack must also incorporate interstitial shielding layers to route high-frequency displacement currents back into the center-tapped decoupling capacitors, reducing common-mode noise generated by the high-dv/dt switching node. Finally, the module must be completed with mechanical housing, adhesive sealing, and external high-current terminals, forming a compact, manufacturable power module architecture consistent with the requirements of the expanded design task.

What Is To Be Found

This project requires determining the key electrical, thermal, and mechanical parameters necessary to design a high density two phase interleaved coupled inductor buck converter. First, the material properties of the Cu/nano-Cu/ARC stack must be evaluated, including equivalent electrical resistivity, thermal conductivity, modulus, CTE, and the resulting stress behavior under operating temperature gradients. These properties are used to determine allowable conductor thicknesses, nano-copper characteristics, dielectric spacing, and the overall feasibility of the multilayer metallization stack.

Next, the electrical performance of the 72 V to 24 V, 1.1 kW converter must be established through SPICE or PLECS simulation at both light load and full load conditions. This includes switching-node waveforms, inductor currents, output voltage behavior, and detailed loss breakdowns for the EPC eGaN FETs. From these results, the required layout and interconnect sizing must be determined, specifically ensuring that the freewheeling current path resistance remains within 3% of the FETs' RDS(on) and that decoupling capacitors, gate drive capacitors, and high-current terminals are placed to minimize loop inductance and conduction loss.

In the expanded phase of the assignment, the project must also determine the requirements for dedicated high side and low side gate driver circuits, including the sizing and placement of gate drive capacitors, bootstrap components, and isolated supply terminals. The design must evaluate how gate driver placement, routing, and parasitic coupling influence switching behavior, turn on/ turn off dynamics, and overall converter stability.

A second major objective is the incorporation of interstitial shielding layers within the ARC substrate stack. These conductive layers must be designed to route high frequency displacement currents back into the center tapped decoupling network, thereby reducing common mode noise generated by the high dv/dt switching node. Determining the optimal dielectric thicknesses, shield layer geometry, and return current paths is essential to achieving effective EMI suppression

The project must also determine the thermal resistance from junction to ambient using ANSYS, including the effect of the selected single-sided Alpha Novatech heatsink, airflow, and device placement. This analysis yields the expected junction temperature at a 70 °C ambient and provides a 2D or 3D thermal profile of the module. Finally, the parasitic inductances and capacitances of critical switching paths must be extracted using ANSYS Q3D. These parasitics are then re-inserted into the electrical simulations to quantify their impact on switching-node overshoot, gate-driver performance, device stress, and overall converter behavior.

Collectively, these tasks define the electrical, thermal, mechanical, and EMI-related parameters required to realize a manufacturable, high-power GaN-based converter module that integrates switching devices, gate drivers, shielding layers, housing, and terminals into a cohesive vertical-stack architecture.

What Is Assumed

The design and analysis of this converter rely on several electrical, thermal, and material assumptions necessary to bound the problem and ensure consistent comparison across simulations. The converter operates at a fixed switching frequency of 200 kHz. For thermal modeling, each EPC eGaN device is assumed to have a junction-to-case thermal resistance of $R_{th\text{-}jc} = 0.2$ °C/W, with the case temperature fixed at 70 °C during full-load operation. Airflow over the heatsink is assumed to be steady and uniform, with the velocity selected to meet the thermal-resistance requirements of the chosen Alpha Novatech heatsink. All thermal simulations assume ideal thermal contact between layers except where explicitly modeled.

Material-level assumptions are also required for the Cu/nano-Cu/ARC stack. The nano-copper bonding layer is treated as a porous copper medium, with its electrical conductivity, thermal conductivity, modulus, Poisson's ratio, and CTE scaled linearly with 30% porosity, as specified in the assignment. The ARC substrate is assumed to behave as a homogeneous ceramic with the provided mechanical and thermal properties, and all failures are assumed to occur within bulk materials rather than at interfaces. Electrically, the GaN FETs are assumed to conduct in reverse when gated on, providing synchronous-rectification behavior with no discrete diode drop. Parasitic elements not explicitly extracted in ANSYS Q3D are assumed to be negligible. These assumptions collectively define the operating environment, material behavior, and boundary conditions used throughout the electrical, thermal, and mechanical analyses.

In the expanded portion of the assignment, additional assumptions are required to support the integration of gate driver circuitry, shielding layers, and mechanical packaging. Each GaN device is assumed to use separate high-side and low-side gate-driver chips, with idealized gate-drive supply terminals providing stable bias voltages to the gate-drive capacitors. Bootstrap components and gate-drive capacitors are assumed to behave ideally unless explicitly modeled, with negligible ESR/ESL except where included in parasitic extraction.

For the EMI-reduction requirements, the interstitial shielding layer(s) embedded within the ARC substrate are assumed to be perfectly conductive, with uniform thickness and continuous coverage except for intentional routing gaps. The dielectric layers separating the shields from the switching planes are assumed to maintain uniform thickness and permittivity, enabling predictable capacitive coupling paths for common-mode current return.

Mechanical assumptions are also introduced for the expanded module structure. The design assumes the use of a single-sided cooling configuration, with the Alpha Novatech heatsink serving as the sole thermal extraction path; no dielectric-fluid cooling or dual-sided thermal interfaces are considered. The adhesive or sealing material used to bond the housings to the ARC substrate is assumed to provide perfect mechanical attachment and electrical isolation, with terminals passing through the sealant without introducing additional parasitic paths. The housing is assumed to be rigid, non-conductive, and thermally insulating unless otherwise specified

Finalized product parameters

Voltage Input	72 V
Voltage Output	24 V
Current Output	46 A
Current Per Phase	23 A
Total System Power Output	1.1 kW
Switching Frequency	200 kHz
Targeted System Efficiency	95%
Maximum Operating temperature	150° C
Maximum Package Dimensions	2.5 x 2.5 in

Material Characterization Approximations

Thickness of Copper Per Layer

Prob-1.1. Most likely your design will use the same tCu across a single layer (e.g. like PCBs) but have different tCu on different layers. Determine tCu for each layer and record values. Also record the nCu calculated characteristics.

Approximated Conductor and Efficiency Losses

Prob-1.2. Determine APPROXIMATE conductor loss and % decrease in converter efficiency. Record your answers.

Estimated Electrical Resistance

Prob-1.3. Calculate “per square” electrical resistance for a Cu/nCu/ARC/nCu/Cu layer (your choice) in your design.

Estimated Thermal Resistance

Prob-1.4. Calculate thermal equivalent resistance for 1D 1mmX1mm area Cu/nCu/ARC/nCu/Cu layer. Record the value.

Material Characterization and Package Design

Layer Stack Description

For the module packaging, a strategy combining an embedded chip power module with a heat spreader is used. The figure below shows the packaging strategy with the layers used to package the GaN devices in the interleaved buck circuit.

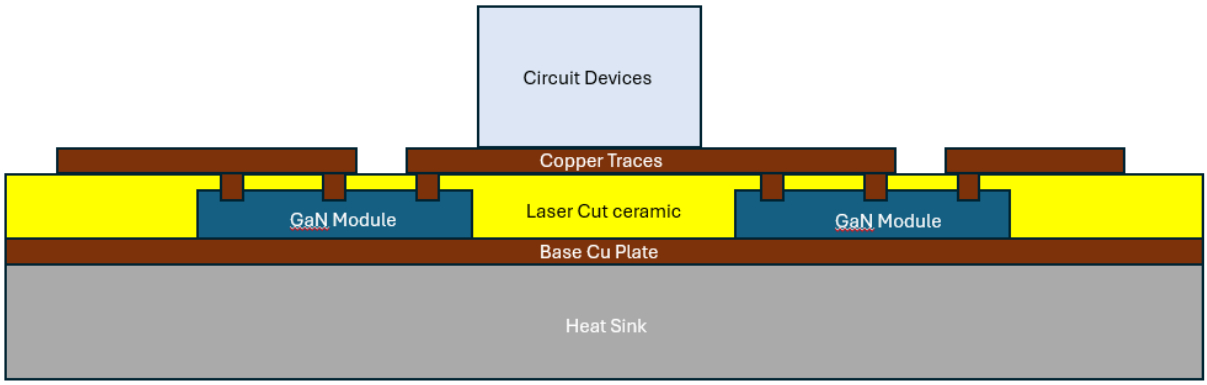


Figure: Cross-sectional layout of the packaging strategy used.

The GaN devices are embedded in a non-conductive ceramic substrate with a high coefficient of thermal expansion, such as alumina (10 x 10^-6/°C) or magnesia (13 x 10^-6/°C). This phenomenon is because the substrate is also directly bonded to the copper traces over it, which helps reduce mechanical stresses. The ceramic substrate is bonded to another copper substrate layer, which is also in contact with the top-cooled GaN module, which helps in thermal conductance directly to the heat sink placed below for heat dissipation

Thermal Considerations -

For thermal considerations, the GaN device chip is assumed to operate at its maximum allowable power loss. This is because the inductive parasitics are too high and may need to be changed in future iterations. However, one can evaluate the package's thermal performance at its thermal limit.

The data sheet for the GaN chip indicates that its highest sustainable operating temperature is 150°C. Typical Rth for non-conductive chips is around 10 degC/W. Hence, taking the ambient temperature as 25°C, the maximum sustainable power loss can be calculated by

$$T = T_{amb} + P * R_{th}$$

As T = 150° C, T_{amb} = 25° C, and R_{th} = 10° C/W, the maximum sustainable power loss is 12.5 watts. This power will be dissipated by each GaN chip through the heat sink; hence, the package must be able to sustain these temperatures.

Mechanical Considerations -

For mechanical stresses induced by thermal expansion, the main failure area is at the layer junctions of the Cu traces over the ceramic layers. As the yield stress of alumina is very high (~2500 MPa), the limiting stresses are the yield stresses of the Cu traces (~320 MPa) and substrate and the heat sink material, AL6061 (~276 MPa). However, as AL6061 is directly bonded to Cu, the main CTE mismatch is between Cu (17e-6/°C) and alumina (10e-6/°C).

Material	**Coefficient of thermal expansion** (/°C)	**Yield stress (MPa)**	**Purpose in Package**
Copper (pure)	18e-6	320	Substrate layer and connections
Al 6061	23.6e-6	276	Heat Sink material
Alumina (88%)	10e-6	2500	Substrate and covering material for chips

Hence, the thermal stresses that are induced due to the CTE mismatch between the alumina and Cu must be less than Cu's yield stress.

Physical Layout (SolidWorks)

In the final package design, the four GaN devices are arranged in a square pattern, with heat generation from four points as far apart as possible to avoid cascading effects. The copper traces are placed according to the GaN source and drain terminals. The coupled inductor is the only other circuit component placed over the copper traces. The capacitors will be added later, once the gate drive is added to the circuit for operation. The copper traces will be edited accordingly. A terminal is added to the drain and source copper pads to provide connection points on the package.

A 3-ounce copper trace is used for the connections between the chips and the coupled inductor. On the underside of the circuit, a heat sink is placed to facilitate heat transfer. The heatsink chosen is Z60-12.7B from Alpha, made from Al6061. As the GaN devices chosen are top-cooled, their thermal plate is in direct

contact with the bottom copper plate, which serves as the heat sink. This provides an efficient path for heat to dissipate through the heat sink.

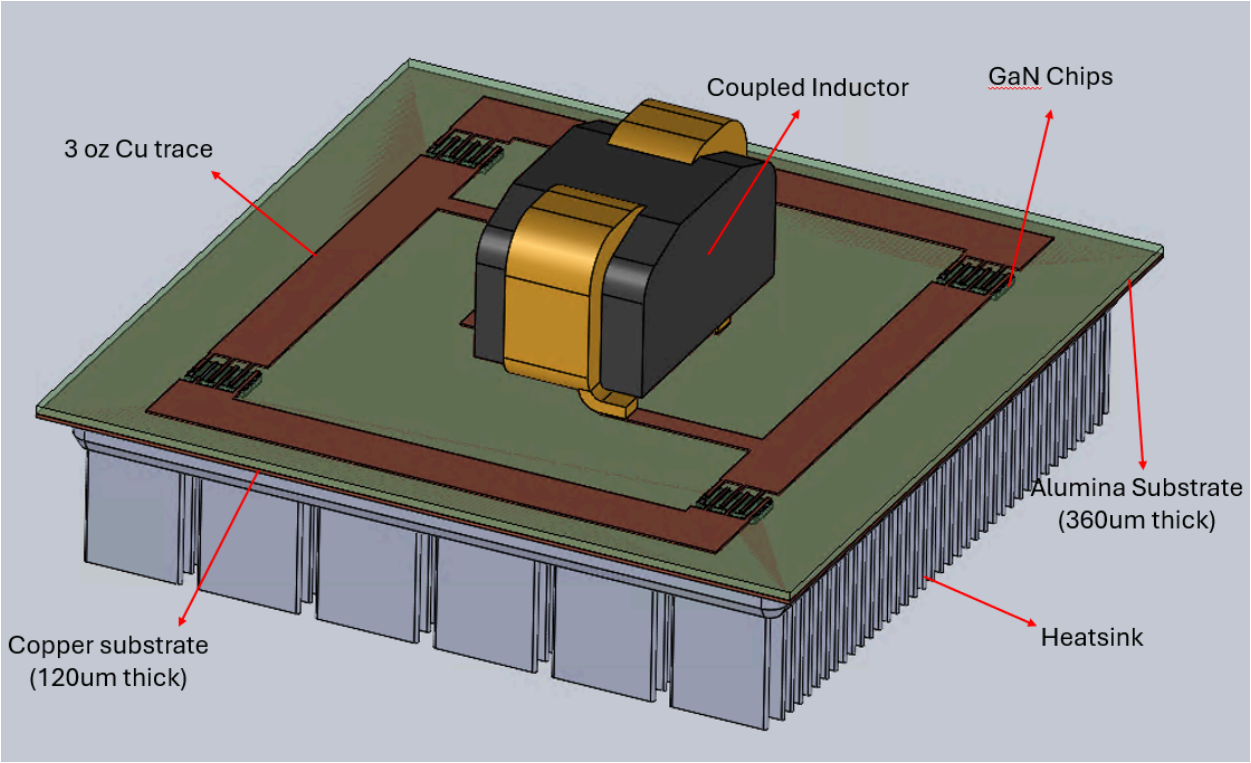


Figure: Package design and components used.

Source and drain terminals were added to the design, and the package was covered with a heat-resistant material attached to the copper substrate to thermally insulate it from all sides except the heatsink, allowing better heat transfer.

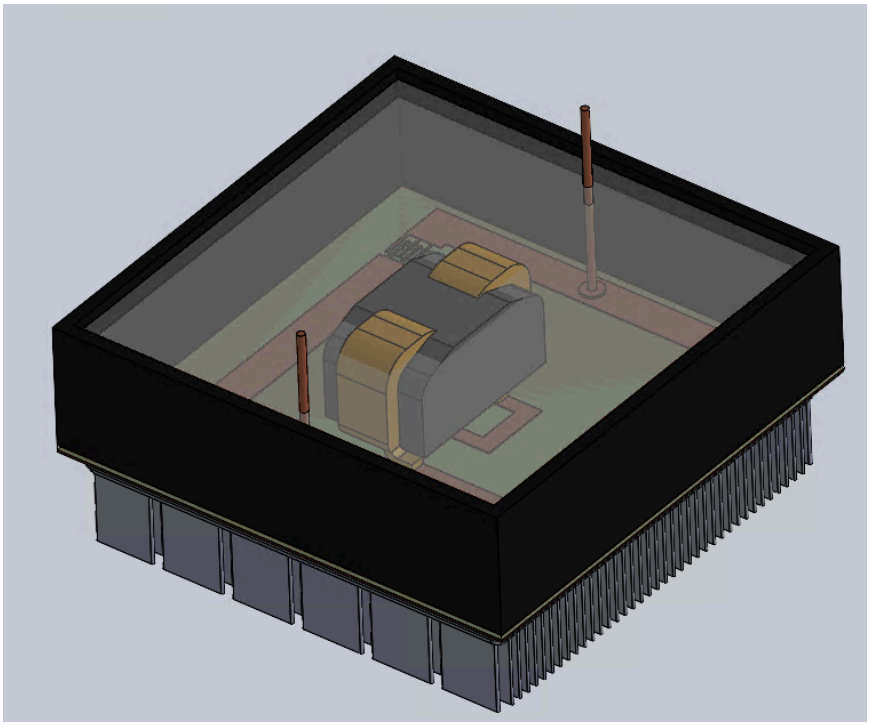


Figure: Final package design with outer cover.

Electrical

Part Selection

EPC GaN MOSFET

Chosen: EPC2302
Backup Option: EPC2304 - this one will be used in case we can't suppress the voltage spikes
Target Resistance: <= 5 mOhm
Required Current Rating: >50A
Required Current Pulse rating: >200A

Going ahead, we may choose to use 2xEPC2304 in parallel

Coupled Inductor

$$L = \frac{(V_{in} - V_{out})DT_s}{(1+k)\Delta I_L}$$

Here,

$V_{in} = 72$ V; $V_{out} = 24$ V; $D = 0.33$; $f_{sw} = 200$ kHz;

$I_{phase} = 23$ A; $\Delta I_L = 5$ A (20% ripple); $k = 0.8$

$$\rightarrow L = 3.05\ \mu\text{H}$$

Chosen Inductor: ERUC-4R1K

The chosen inductor value is slightly larger than the calculated value due to the required Current rating and options available from TDK.

Input Capacitor

Ceramic caps: Roughly 40uF target. To reduce the ESR effect, we will use 5x10uF rated at about 150V, accounting for 20% losses.

Model: GRM32EC72A106KE05L x5

Electrolytic caps: Roughly 0.4 mF. Will consider using 4x100 uF electrolytic caps, rated about 150 V, accounting for 50% oversizing

Model: EEU-EE2C101 x4

Input capacitor: Ceramics for ripple filtering, electrolytic for energy storage.

Ceramics $\rightarrow$

$$\Delta V_{in,p-p} = \frac{I_o D(1-D)}{f_s C_{in}} \rightarrow C_{in} \approx 34\ \mu\text{F}$$

Electrolytic $\rightarrow$

$$\text{Energy} = V_o I t = 24 \cdot 23 \cdot 50 \times 10^{-6} = 0.0276\ \text{J}$$

$$0.0276 = \frac{1}{2} C \left(72^2 - 70.5^2\right) \rightarrow C = 0.39\ \text{mF}$$

Gate Driver

Bootstrap Capacitor

Electrical Analysis (LT SPICE)

Ideal Scenario Schematic (No parasitics)

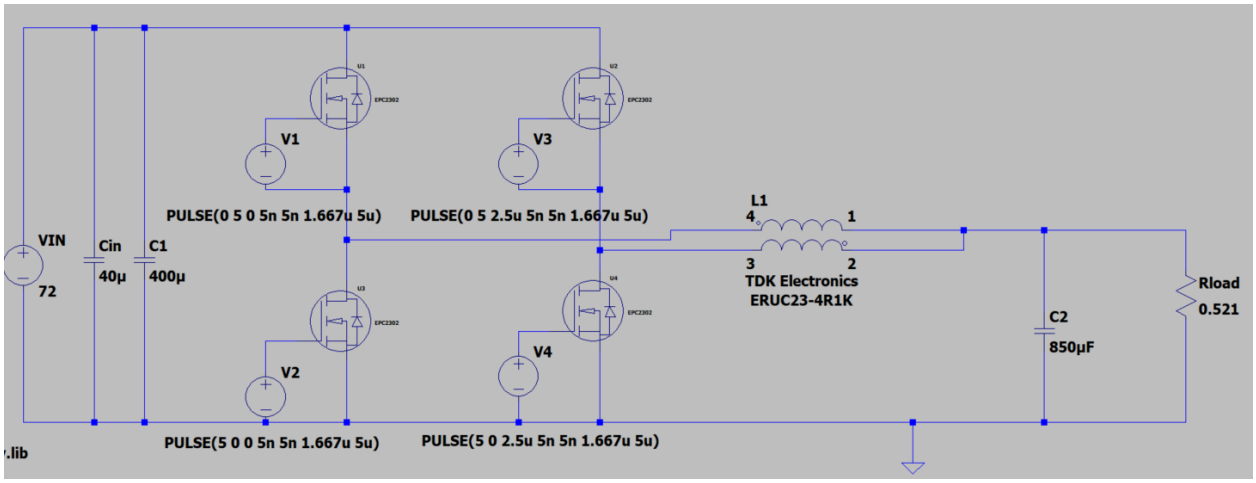


Figure: Ideal Circuit Schematic

Operating Waveforms

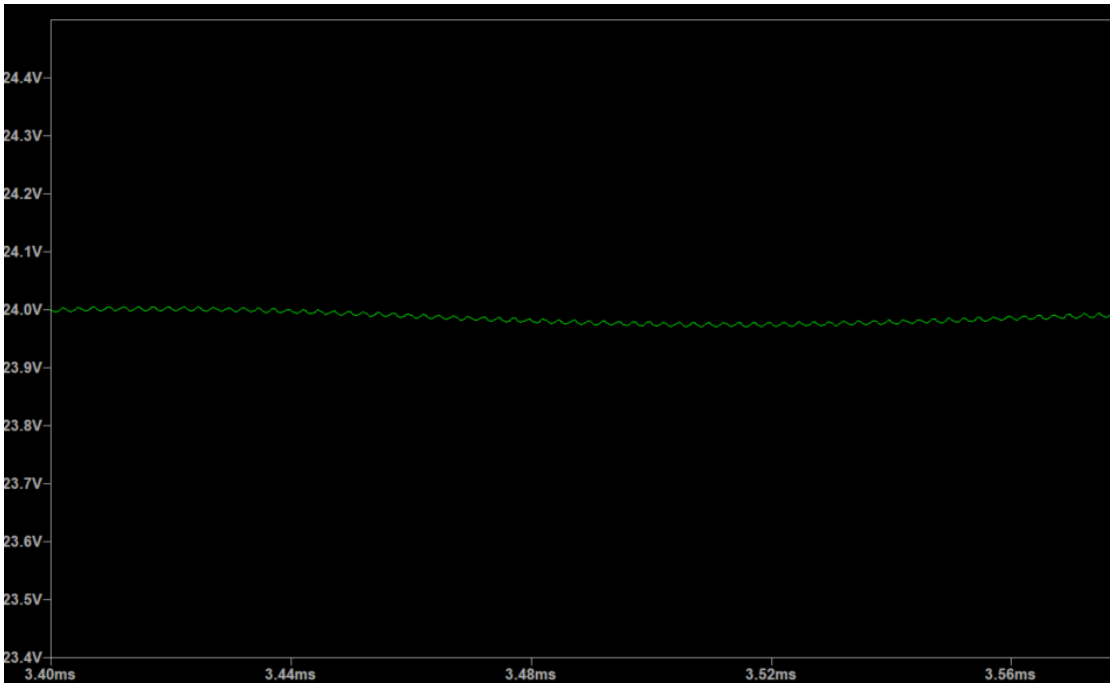


Figure: Output Voltage simulated ~24V

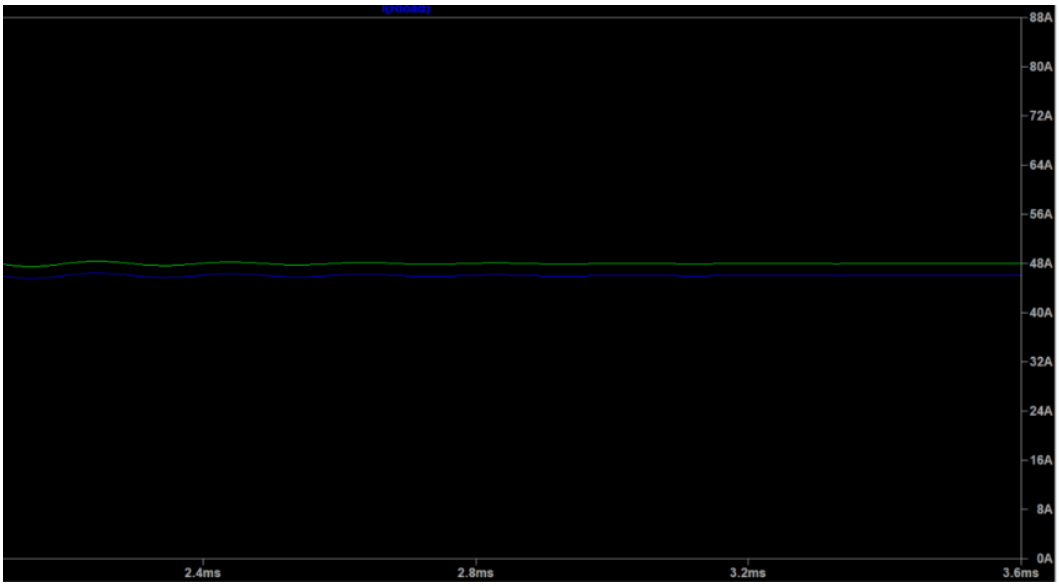


Figure: Output Current Simulated ~46A

MOSFET Switching (V_DS) waveform

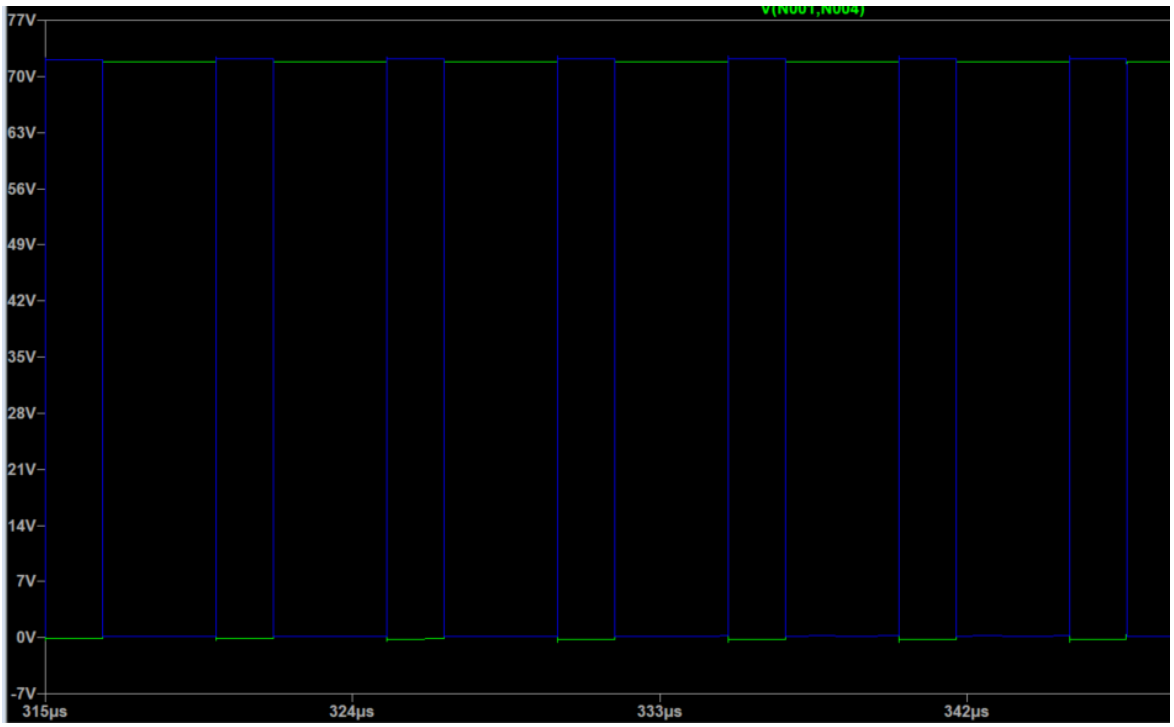


Figure: V_{DS} Switching across Half-bridge MOSFETS

Phase output waveform (Current and Voltage)

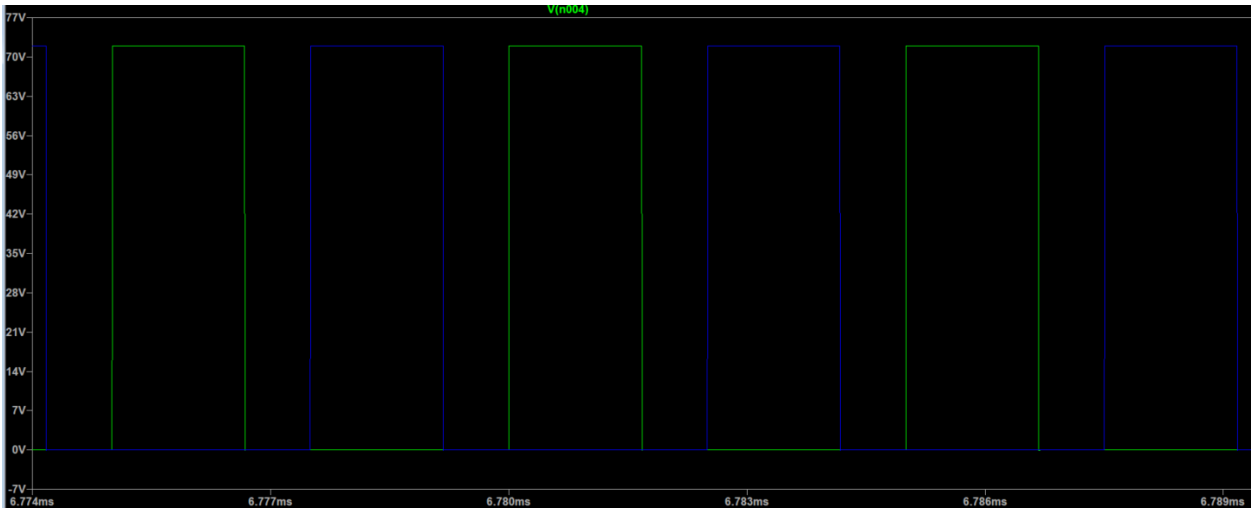


Figure: Voltage Waveform for Half-bridge Outputs

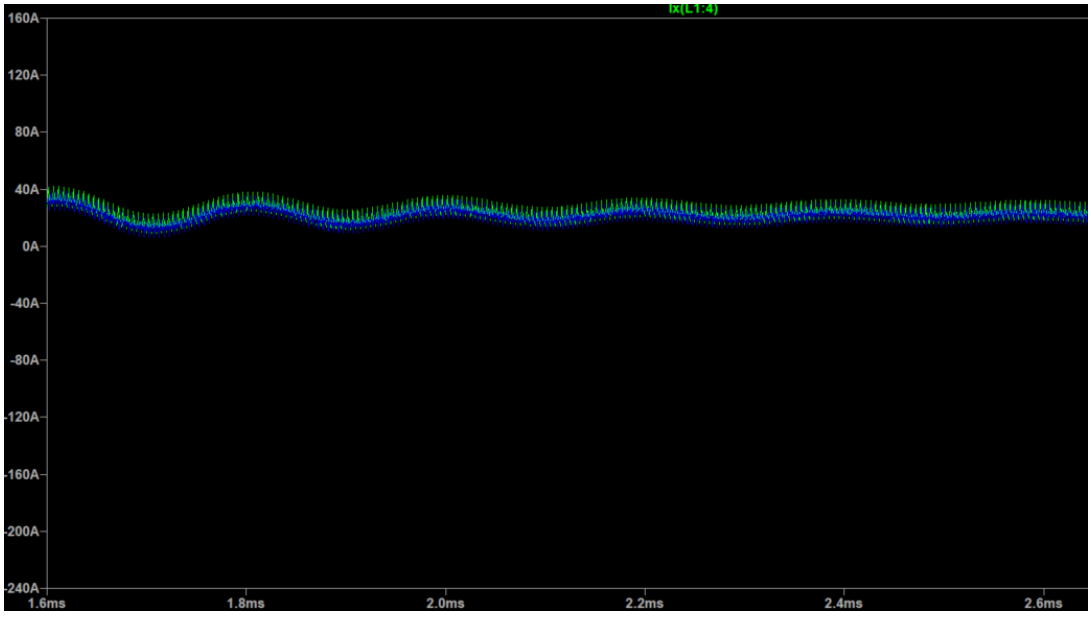


Figure: Current Output on phases ~23A

Loss Breakdown

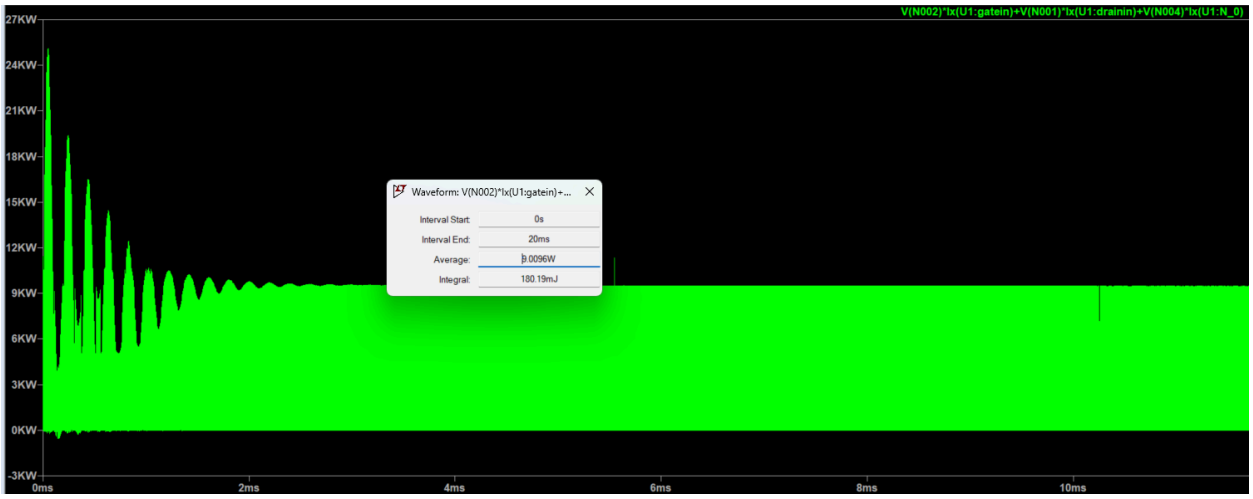
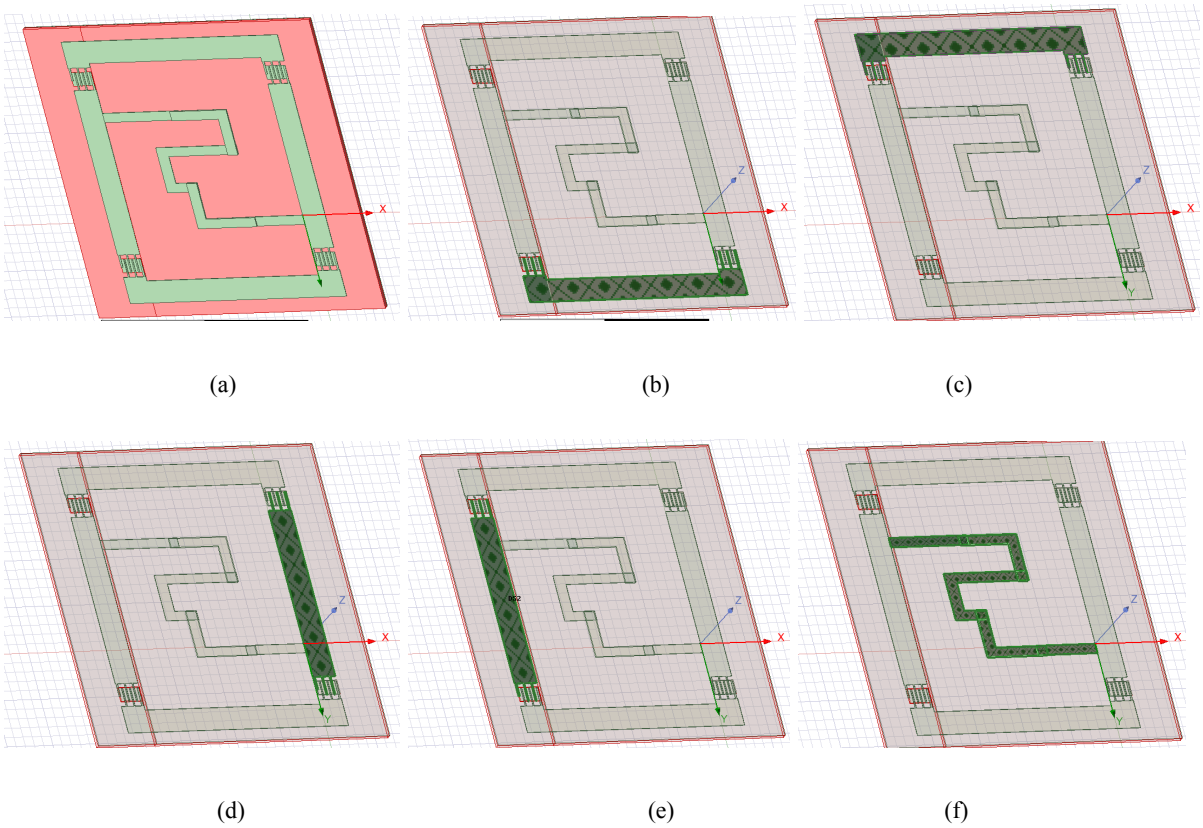


Figure: Power Loss across a MOSFET ~9W

Parasitic Extraction (ANSYS Q3D)

Critical Loop Identification



(a) (b) (c)

(d) (e) (f)

Figure: (a) Q3D model; (b) DC+ net; (c) DC- net; (d), (e) Device interconnection net; (f) Inductor loop net

The figure above shows the Q3D model and the defined nets. Inductor pads are shorted with copper traces to reduce computational burden. In total, 5 nets are defined, one for the path between the drains of the top devices (DC+), one for the path between the sources of the bottom devices (DC-), two for the paths between the upper and lower devices, and the last one for the loop containing the inductors (it starts and ends on the switch nodes). The inductor loop is not as critical as the other ones, because its parasitic inductances would add up to the main inductance value. However, the parasitic inductance of the other nets will directly affect device switching.

Inductance Extraction

The solution frequency is set to 2 MHz, which is ten times the switching frequency. Excitation is set to a constant voltage because all the sources/sinks of the nets are connected to some physical component. The next figure shows the DC and AC parasitic resistance and inductance in Ω and nH, respectively. AC results are more relevant to the analysis because they are calculated at high frequencies. As shown, parasitic inductance in the inductor loop is the highest due to the trace length. However, this is an

over-estimation of the actual values, because inductors are shorted with copper traces, which also adds parasitic inductance. Nevertheless, this is acceptable, as the parasitic inductance for this net will be added to the main inductance and will not have a major effect on switching. The inductance of DC+ and DC- nets is very small because the current path in these nets is short and wide, which reduces parasitics. The inductance of the interconnection nets is 10.8 nH, which will significantly affect device switching. The reason for this high value is that the traces here are narrow and almost 36 mm long.

	Self
Freq: 2MHz	
BossExtrude22:Source2	0.00006, 0.82874
BossExtrude24:Source1	0.00006, 0.82910
DS1:Source3	0.00105, 23.10736
DS2:Source4	0.00105, 23.11399
Lpath:Source5	0.00449, 55.41178

(a)

	Self
Freq: 2MHz	
BossExtrude22:Source2	0.00020, 0.54639
BossExtrude24:Source1	0.00018, 0.54435
DS1:Source3	0.01028, 10.80934
DS2:Source4	0.00958, 10.82719
Lpath:Source5	0.01785, 25.35079

(b)

Figure: (a) DC RL results; (b) AC RL results

Capacitance Extraction

	Self
Freq: 2MHz	
BossExtrude22	0.00385, 18.41631
BossExtrude24	0.00384, 18.32120
DS1	0.00293, 16.26812
DS2	0.00295, 16.35639
Lpath	0.00243, 18.22818

Figure: CG results

The figure shows the parasitic capacitances (in pF) of all the nets with respect to the heatsink. All the nets have the same capacitance because of uniform separation. The reason the interconnection nets show a slightly lower value is that, to separate them from the inductor loop, a small gap was inserted into the Q3D model. This gap causes some capacitive coupling between the nets, so all nets can be assumed to have the same capacitance with respect to the heatsink.

Updated Electrical Simulation

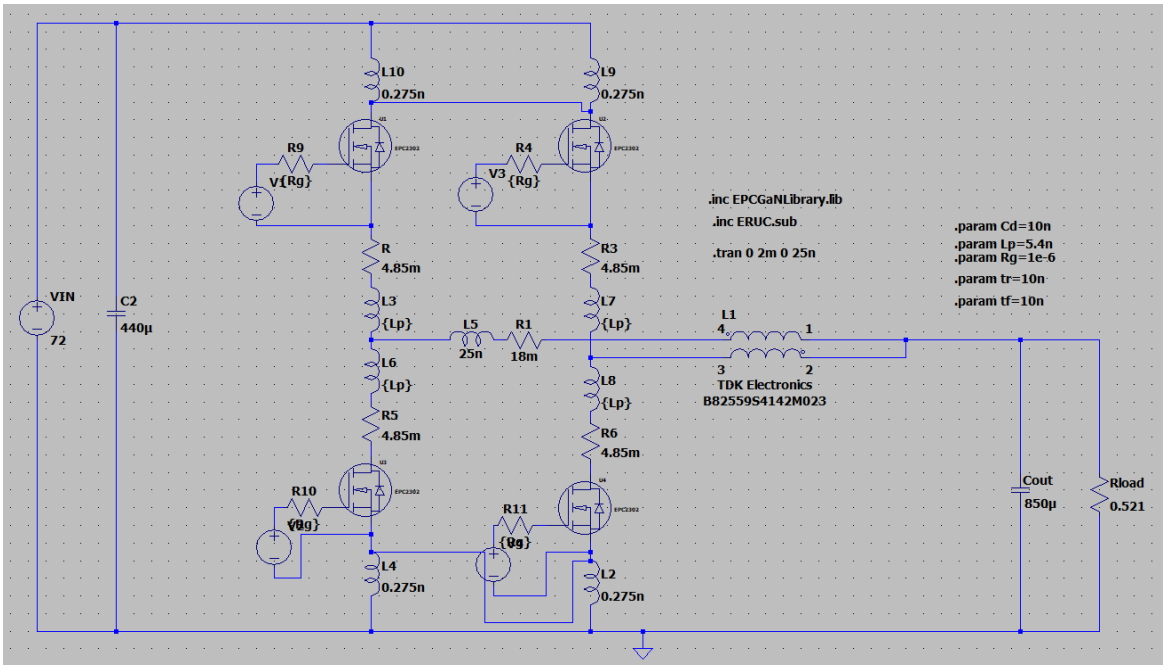
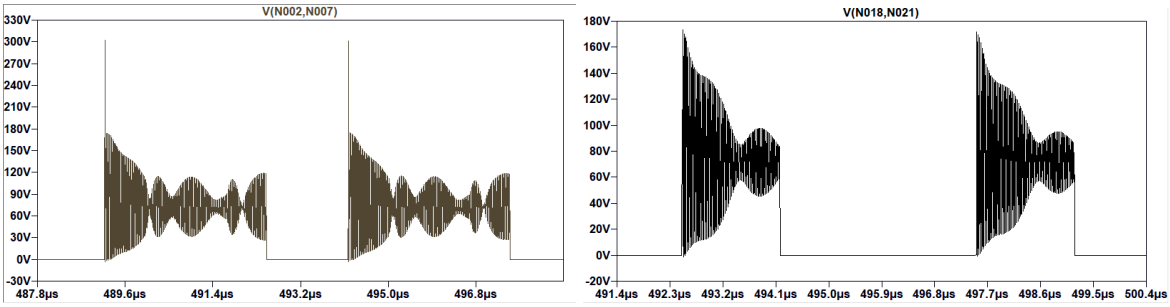


Figure: Spice model with parasitics

The parasites obtained from Q3D are included in the spice model, as shown in the figure above. The rise and fall times of the pulse sources are set to 10 ns, and the gate resistance is set to a negligible value to observe switching during gate pulses with a very high slew rate.



(a) (b)

Figure: V_{ds} waveform for (a) top devices; (b) bottom devices

As shown in the waveforms, the top device experiences overshoot beyond its rating, which is destructive to this device. In terms of loss, all devices see dissipation below 5 W at this point. To tackle overshoot and ringing, one approach would be to redesign the geometrical layout, ensuring that interconnect nets are as short as possible so that parasitic inductance drops to a very low value, such as 1-2 nH. Another approach would be to decrease the slew rate of the gate pulses (by changing the gate resistance), add decoupling capacitors, and add snubbers. The second approach is taken in the mid-project to investigate its

drawbacks. The next figure shows the modified Spice model after applying these solutions. The gate resistance for turn-on is set to 5 Ω, and for turn-off to 15 Ω. The snubber resistance is 3.3 Ω, and the capacitance is 220 pF.

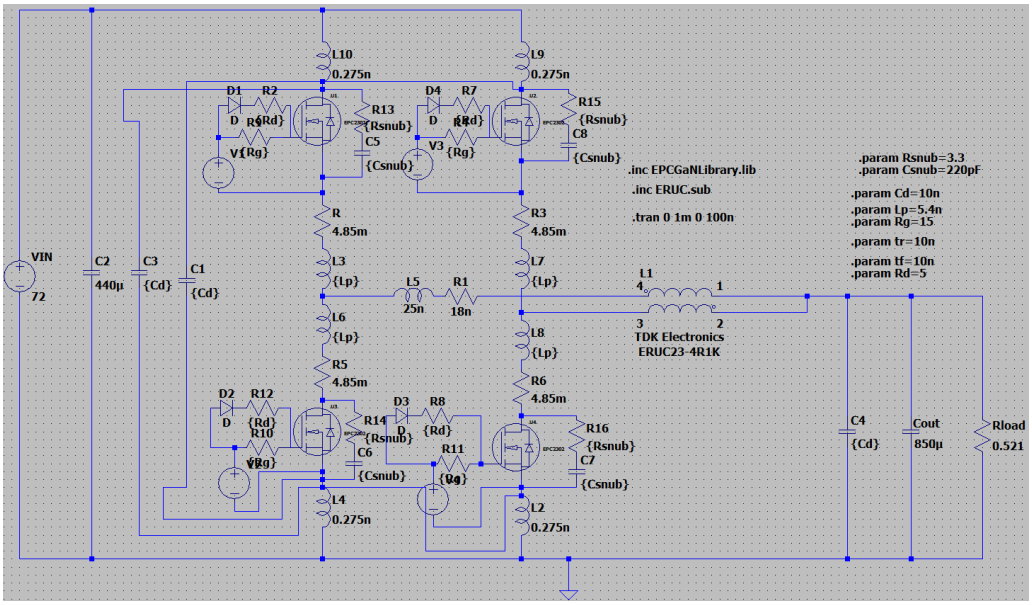
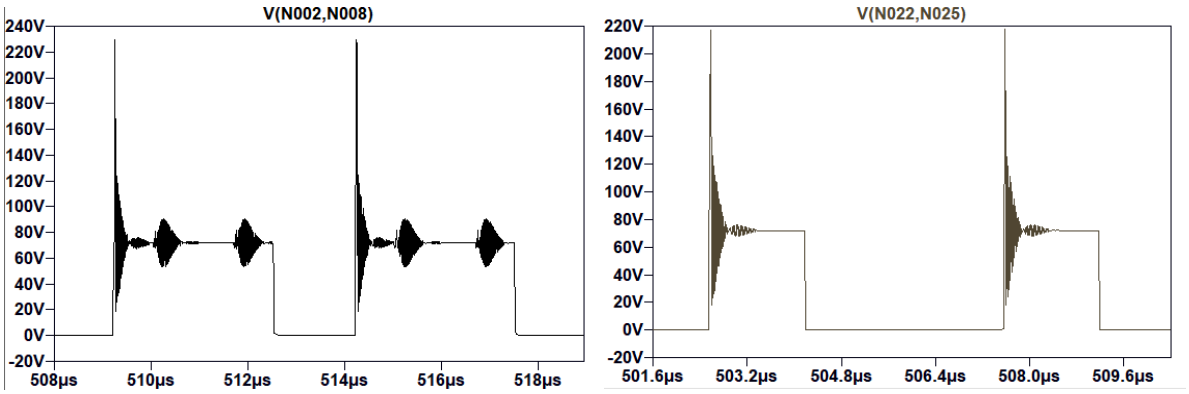


Figure: Modified Spice model

The next figures show the V_{ds} waveforms.



(a) (b)

Figure: V_{ds} waveform for (a) top devices; (b) bottom devices

Thermal Analysis (ANSYS)

The thermal simulation is carried out in ANSYS Workbench's static thermal, which uses an APDL FEA solver. For the workbench simulation, the model is updated by replacing the GaN chips with a similar design made out of a conductor (AL6061) to simulate most heat transfer from the chip (designing for the

worst-case scenario), with a conductive plate at the top made out of AL-6061, which is part of the top-cooled thermal chip.

The simulation materials are assigned from the ANSYS GRANTA library according to the design in Table 1. The coupled inductor is not included in the thermal simulation, as the effect of thermal expansion is to be investigated on Copper traces. The inductor would actually help the copper traces from expanding, thereby helping the package stay better secured, as the alumina substrate has a lower CTE.
For the simulation, two sets of designs are considered: the package without the heat sink and the package with the heat sink. For both simulations, it is assumed that the package is air-cooled with forced convection using a fan with an air velocity of 8 m/s, which can be easily achieved with a fan for a package size of 2.5 in x 2.5 in.

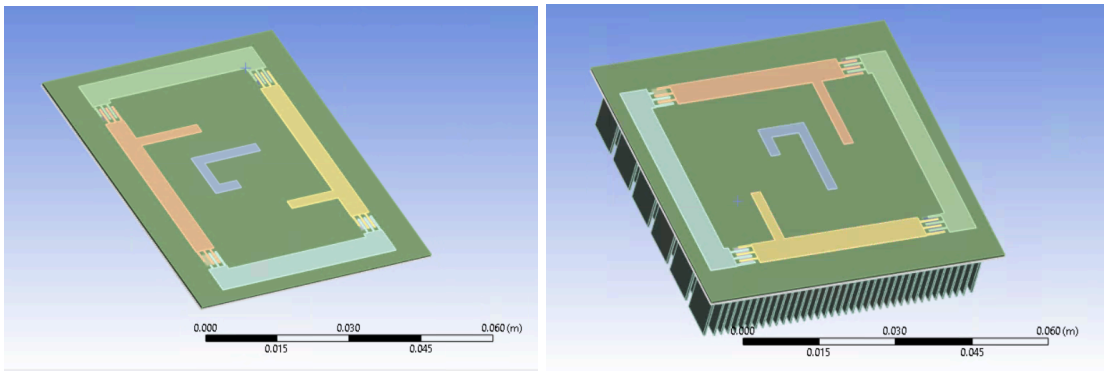


Figure: Geometry used for thermal simulation for the package without a heatsink (left) and with a heatsink (right).

Simulation Setup

The model is meshed with tetrahedral elements for the copper traces and the heat sink, and with hexahedral elements for all other parts. A mesh size of 0.12mm is applied to the copper traces to improve solution quality.

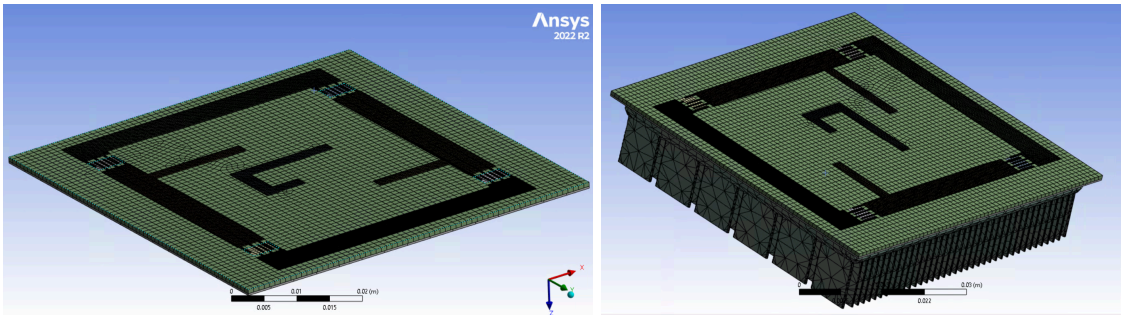


Figure: Mesh used for thermal simulation for the package without heatsink (left) and with heatsink (right).

For the thermal analysis, a temperature of 150° C is applied to the top surfaces of all the GaN chips, as previously mentioned. A convection boundary condition is applied to the bottom copper plate for the non-heat sink design, whereas for the heat sink design, all exposed surfaces are given a convection boundary condition.

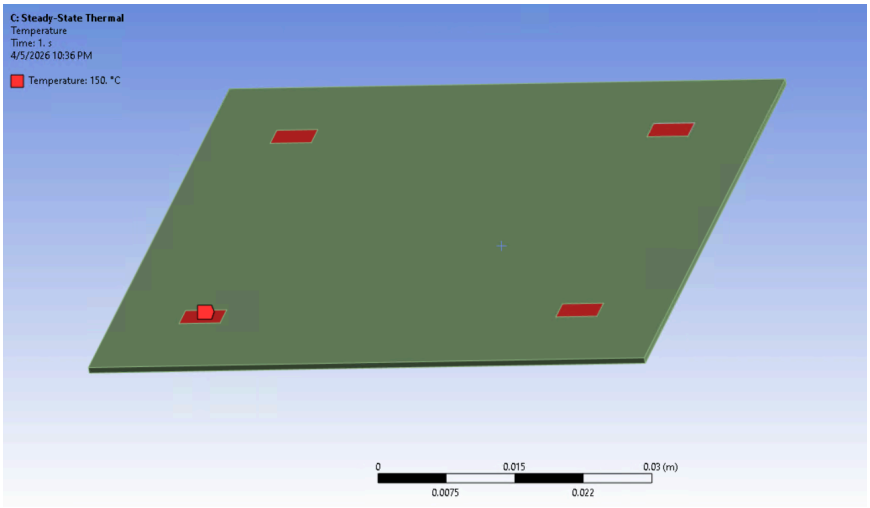


Figure: Temperature boundary condition of 150° C applied at the top planes of the GaN chip (top cooled package) for both the simulations.

Now, assuming a fan speed of 8m/s the convective heat transfer coefficient for a copper plate can be calculated as follows -

Air velocity, $V = 8$ m/s
Plate length, $L = 0.0635$ m
Kinematic viscosity of air, $v = 1.5 \times 10^{-5}$ m²/s
Thermal conductivity of air, $k = 0.026$ W/m·K
Prandtl number, $Pr = 0.71$

Reynolds number:
$Re = (V \times L) / v$
$Re = 3.38 \times 10^4$

Since Re < 5 × 10^5 → flow is laminar

Nusselt number (laminar flow over flat plate):
$Nu = 0.664 \times Re^{1/2} \times Pr^{1/3}$
$Nu \approx 109$

Convective heat transfer coefficient:

$h = (k \times Nu) / L$

$h \approx 44.6\ W/m^2 \cdot K$

Thus, a convection coefficient of 45 W/sq-mK was used for the non-heat-sink simulation. For the heat sink simulation, the convection coefficient could not be accurately calculated because the characteristic length of the heat sink was not provided in the heat sink data sheet. However, generally heatsinks increase the convection coefficient by a factor of 1.5-1.6. Hence, a conservative value of 80W/sq-mK is taken for the convection simulation with a heat sink.

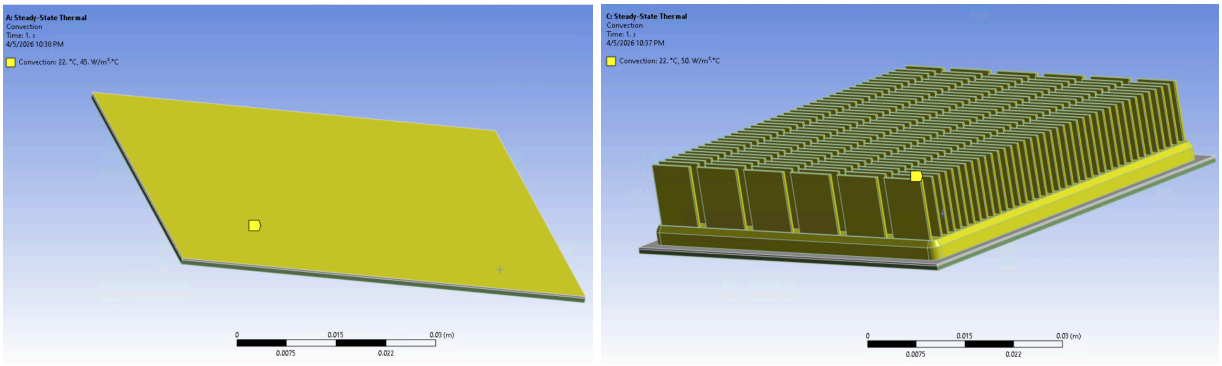


Figure: Convection planes and coefficient of convection used for thermal simulation for the package without heatsink (left) and with heatsink (right).

Thermal Simulation Results

The temperature distribution for the two simulations is shown in the figures below.

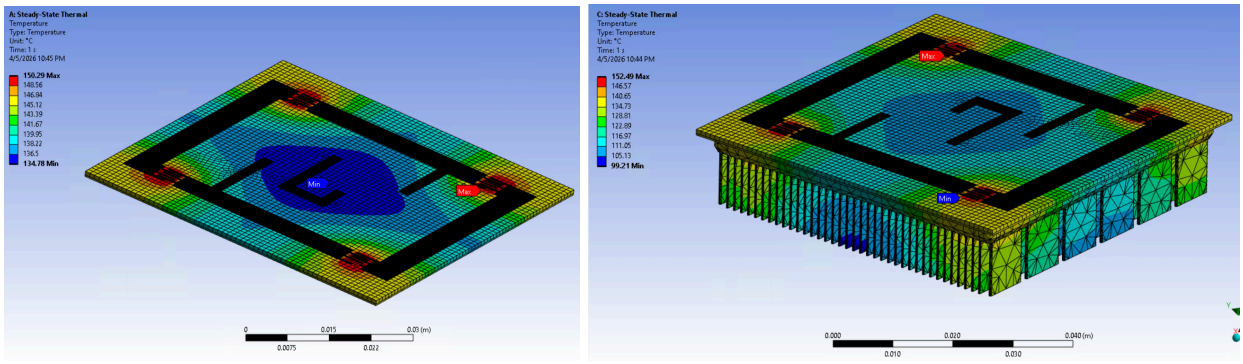


Figure: Temperature distribution observed for the package without heatsink (left) and with heatsink (right)

It is observed that, without the heat sink, the package temperature remains between 153° C and 134° C, with the highest temperatures near the GaN chips and the lowest at the center of the package.

For the simulation with the heat sink, the temperature range is from 153° C to about 100° C, with a temperature drop of about 50° C (20° C more than the on heat sink package). The lowest temperatures, however, are observed on the heat sink, with 110° C recorded in the mid-section of the package.

Mechanical Analysis (ANSYS)

With the applied thermal loads, mechanical stresses will be induced by thermal expansion and material mismatches. This can potentially lead to material failure and lead to the connectors peeling off the package.

To analyze this, the thermal results for the heat sink and the non-heat sink simulations are transferred to the static structural solver in ANSYS Workbench, which is an implicit solver used for constant loading conditions that do not vary much with time (static or quasi-static).

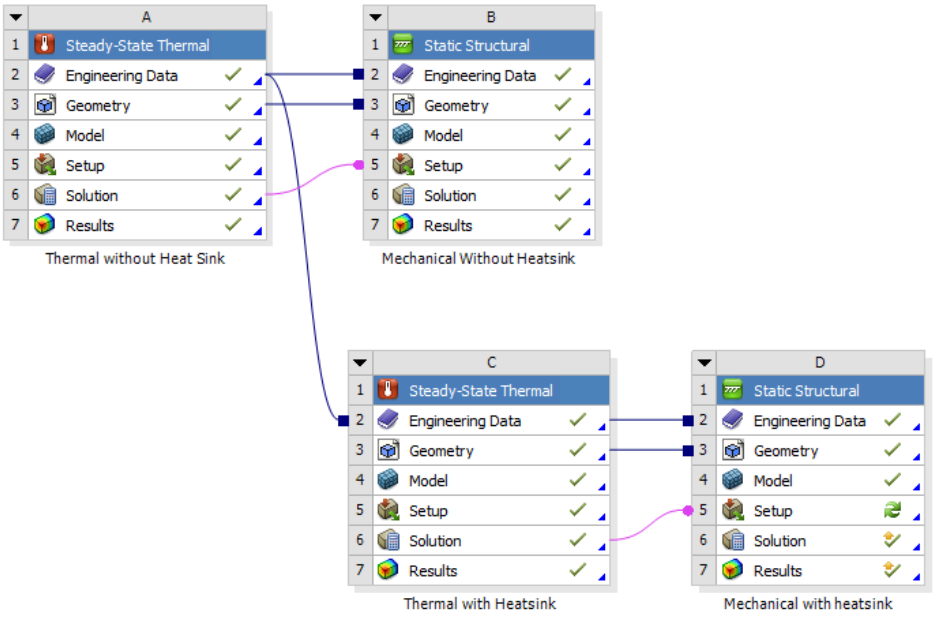


Figure: Thermal load transfer for mechanical simulation.

Simulation Setup

The meshing is done in a similar way to the thermal simulation. All the inter-body connections or contact faces are set to the 'bonded' condition, as there is to be no sliding in between the layers of the package.

For the boundary conditions, the thermal load from the thermal simulation is applied. Moreover, to constrain the package in space, three mutually perpendicular frictionless support planes are added, which

help prevent the package from expanding. This ensures the package does not incur additional loads due to boundaries.

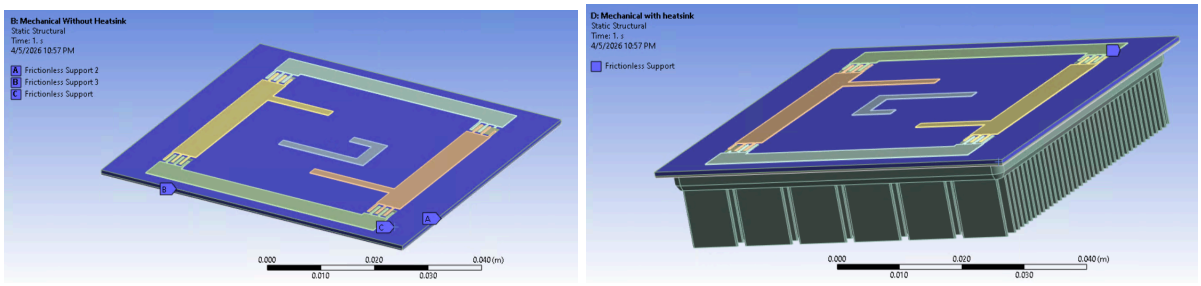


Figure: 3 mutually perpendicular frictionless planes chosen as boundary conditions to constrain the package in space for both the simulations.

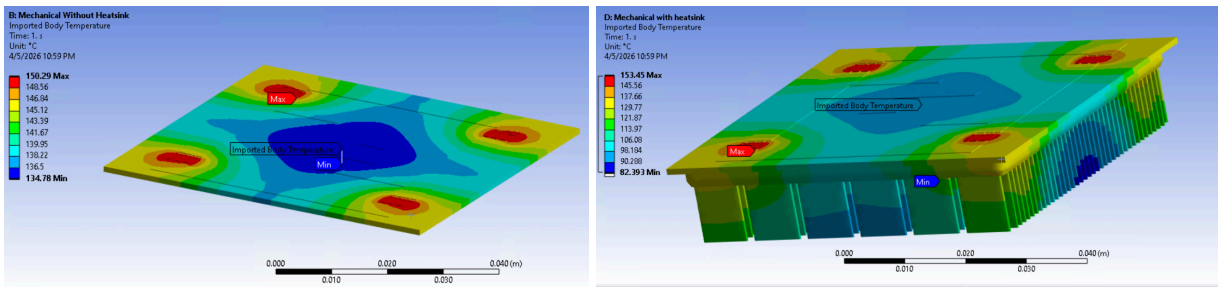


Figure: Imported temperature loads from the thermal simulation for both the models..

Mechanical simulation results

Using the above boundary conditions and constraints, the models were simulated for stresses and total deformation (i.e., expansion).

It is observed that the maximum stress for the package without a heatsink reaches 6626.6 MPa.While the stress in the other areas of the package is within the safe region, this stress is observed in a small area at the junction between the GaN chips and the copper traces. However, this stress is too high and exceeds the yield stress of copper, which is 320 MPa. This could lead to potential disconnection due to peeling of copper traces. Total deformation is also observed to be 0.11 mm, which is actually in the favorable range for a package measuring 2.5 in x 2.5 in.

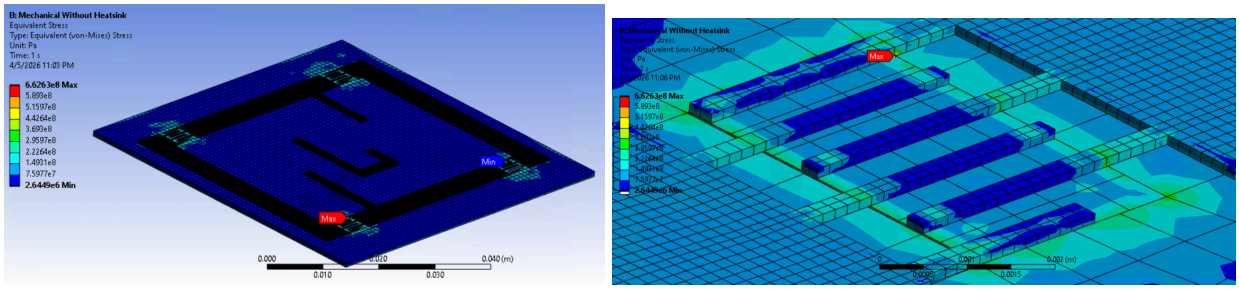


Figure: Equivalent von-Mises stress distribution on the package without heatsink (left) and point of maximum stress of 6626.6MPa (right).

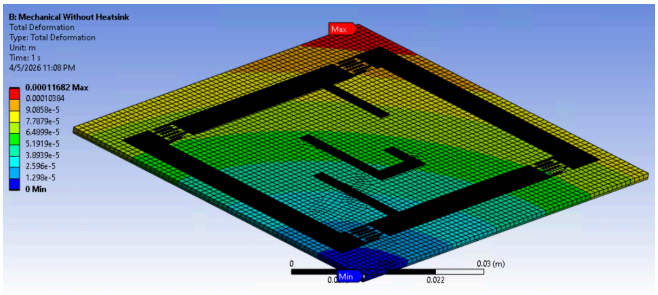


Figure: Total deformation observed in the package without heatsink.

With the heatsink, the maximum stress observed is 282 MPa, which is lower than the copper yield stress. Moreover, the majority of high stress is observed at the periphery of the package near the edges of the copper substrate, which is the best place, as it is far from the copper traces, which are at a lower stress. Hence, there is no chance of the copper layer peeling off in the package, making it suitable for operation with a power loss of 12.5 W or a temperature rise of 150°C.

The deformation of the package itself has not changed much with the addition of a heat sink, with a maximum deformation of 0.14 mm, similar to the value observed in the simulation without a heat sink.

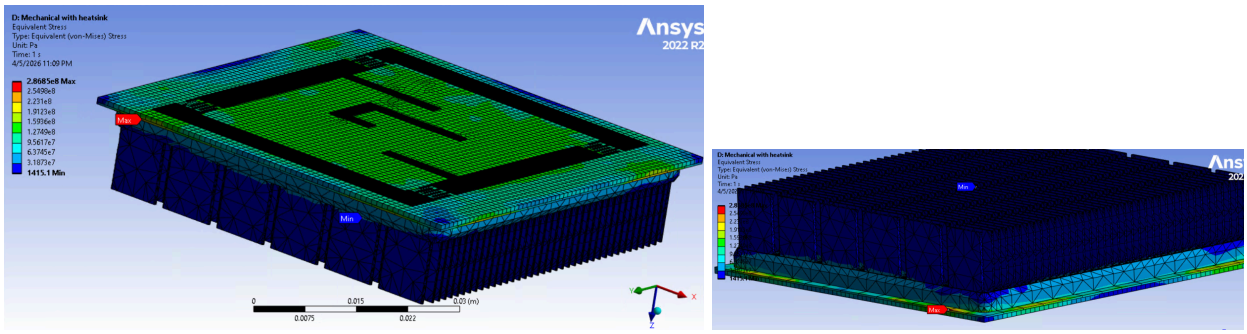


Figure: Equivalent von-Mises stress distribution on the package with heatsink (left) and point of maximum stress of 282MPa (right).

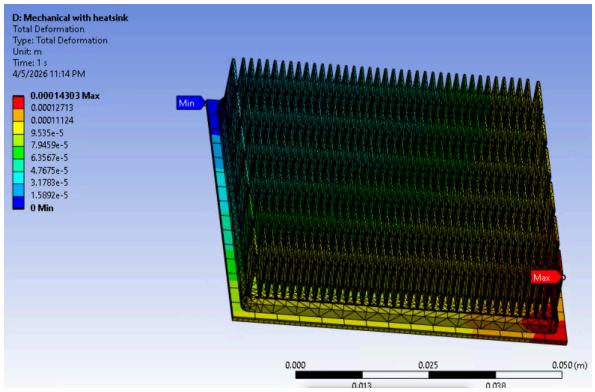


Figure: Total deformation observed in the package with heatsink.

Conclusions

Our design is not optimal due to the high loop inductance and parasitics. This issue causes a large voltage overshoot across the drain and source of the MOSFETs, which would destroy them. To address this, we have used a gate resistor and an RC snubber to electrically dampen the effect, but this results in a thermally inefficient/impractical design. While this may not be a functional design, our initial electrical simulations have shown that with minimal parasitics, the product targets can be achieved with some fixes.

That said, the package was tested to a maximum high-temperature limit of 150°C, as rated by the chip manufacturer. The package without a heatsink itself showed high stresses near the copper trace junctions with the chips when the GaNs were operated at 150°C. However, with a heatsink, the package can withstand the thermal load without any failure of the packaging materials.

Changes to the MOSFET package, the operating range, and the inductor copper connectors would be needed to make the package operation feasible as an interleaved buck converter.

Further Improvements

We plan to revise our design to mitigate parasitics, voltage overshoot, and thermal efficiency, and to implement better packaging with gate control in the next milestone.

- To achieve this, we are considering switching to a 2- or 3-parallel MOSFET design with the EPC2304 GaN MOSFET.
- Along with this, we plan to design a better package with lower loop inductance and parasitics to prevent the overshoots.
- The RC Snubber and Gate resistors will also be more tuned in the next version.

References

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